Investigation into the effects of emergency spray on thermal runaway propagation within Lithium-ion batteries

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Abstract： Inhibition a battery's thermal runaway propagation can avoid serious accidents in electric vehicles. Emergency spray has been proven to be effective in suppressing thermal runaway of a single cell, but the effect of inhibiting thermal runaway propagation between multiple batteries still needs further investigation. In this study, the characteristics of thermal runaway propagation was experimentally investigated, and the emergency spray technology, with different cooling durations, was applied in various stages of thermal runaway propagation. The results indicated that the continuous spray could not only reduce the average maximum temperature, but also delay the diffusion among multiple batteries and provide further response time. However, after-combustion was a frequent occurrence when the spray shut down. A correlation model was proposed to evaluated the maximum heat production of lithium-ion batteries by calculating the enthalpy and mass of the reactants and short circuit energy. The required cooling quantity is determined by experiment and verified by experiment. These results could help the development of a cooling strategy for suppressing the thermal runaway propagation effectively with minimum cooling capacity.

Keywords: Thermal runaway propagation; Spray cooling; Internal short circuit;

Quality loss

1 Introduction

The lithium-ion batteries are quickly becoming a critical part of energy storage systems, including in electric vehicles, because of their superior energy storage performance. However, in light of recent events about lithium-ion batteries, such as the explosion of energy storage power stations and the spontaneous combustion of electric vehicles, it is becoming extremely difficult to ignore the existence of safety issues with the use of lithium-ion batteries. The main cause of these accidents is the thermal runaway in batteries. Current research shows that when the batteries under intense conditions, there is a chain reaction within the batteries, the heat will build up quickly, and eventually catch fire. This is called battery thermal runaway. The amount of heat generated during that time is sufficient to induce thermal runaway in neighboring cells. Battery packs can contain hundreds or even thousands of series-parallel connected cells. Once thermal runaway spreads through the whole battery pack, the damage is enormous. Thus, there is an urgent need to address the safety issue caused by the thermal runaway.

Past studies have provided important information about the mechanism and propagation mode of battery thermal runaway. There are three main trigger inputs of battery thermal runaway: mechanical abuse, electrical abuse, and thermal abuse, and all are related to each other[1]. Studies, such as that conducted by Dubaniewicz et al. [2], Mao et al. [3], Feng et al. [4], and Liu et al. [5], have shown that mechanical abuse can contribute to the structural damage of the battery, cause internal short circuits and generate a large amount of heat, which may eventually lead to thermal runaway. Christian et al. [6] and C. Lin et al. [7] found that electrical abuse also can produce a significant amount of heat and R. Spotnitz et al. [8] concluded that this heat is likely to lead to thermal runaway. Electrical abuse can also destroy battery structure [9]. In addition to mechanical abuse and electrical abuse, contact loss of connectors can also lead to thermal abuse [10], which can trigger thermal runaway of a lithium-ion battery.

Thermal runaway can quickly propagate from one battery to an adjacent battery. There are many factors that affect thermal runaway propagation. Zhang et al. [11] proposed the frontier concept of thermal runaway propagation and the velocity of thermal runaway is correlated to the thermal conductivity and heating rate of the battery. Wang et al. [12] identified that the thermal runaway propagation rate increases with an increase in the diameter of the lithium-ion battery. Liu et al. [13] found that the thermal runaway propagation rate decreased with the reduction of ambient pressure. Li et al. [14] determined that the higher the states of charge, the faster the thermal runaway propagate. There are two different propagation fronts, regular polygons and circles, observed in Jia et al.'s experiment [15]. Li et al. [16] proposed that heating power affected the propagation and transfer mode of thermal runaway. In order to restrain thermal runaway propagation, researchers have proposed a range protective measure.

Ouyang et al.[17] installed the aluminum alloy side plates with high thermal conductivity on one side of the battery. They pointed out that the side plates can delay the onset time of battery thermal runaway and reduce the time interval of thermal runaway propagation. Weng et al. [18] weakened the combustion and extended the propagation time by reducing the oxygen concentration. Rui et al. [19] summarized the critical conditions for thermal runaway propagation by combining insulation with the cooling plate. Many inhibitors such as HFC-227ea [20], C6F12O [21], and Novec 1230 [22] have also been studied in order to inhibit thermal runaway propagation. The inhibitory effect of these inhibitors is relatively modest, and more inhibitors need to be proposed and investigated.

As an effective cooling method, emergency spray has good thermal uniformity and high working fluid utilization, which has attracted increasing attention [23]. It was proved by Saw that spray cooling has good heat dissipation properties, and spray cooling may be a solution for the thermal management of lithium-ion battery packs [24]. Emergency spray technology has been proven by Huang to be effective in suppressing the thermal runaway of an individual cell [25]. Considering the battery pack is composed by a significant number of cells, the emergency spray in actual use would not be able to focus precisely on the individual thermal runaway cell. The effects of spray cooling on the spread of thermal runaway among batteries requires further understanding.

To address this gap, the process of thermal runaway propagation is discussed in this study. The effect of spray cooling on the thermal runaway propagation process is studied by controlling spray time. The results could provide a reference for methods to inhibit thermal runaway propagation.

2. Methodologies

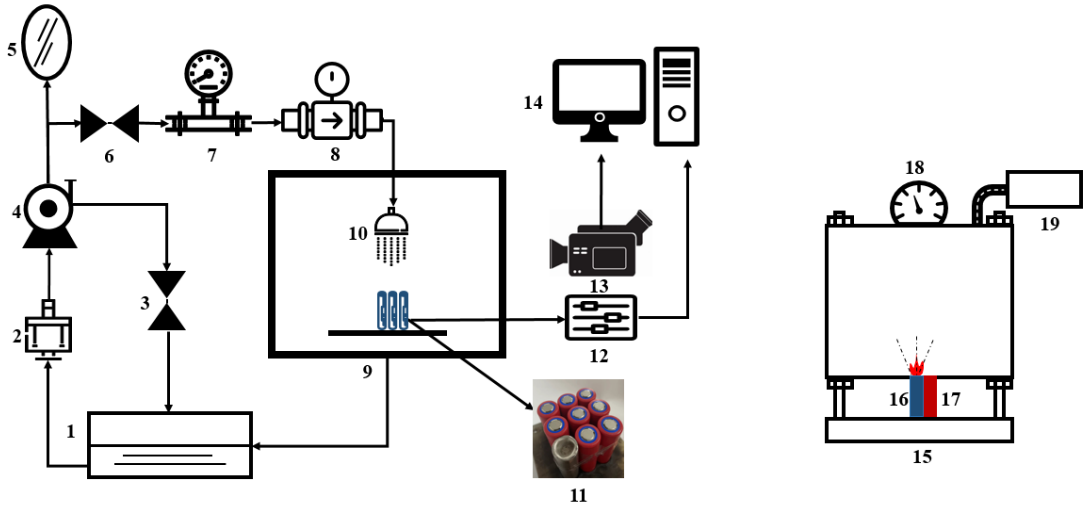
2.1 The experimental apparatus

The 18650-type lithium-ion batteries (SANYO NCR18650G-H09NA),which consist of $Li(Ni_xCo_yAl_y)O_2$ cathodes, carbon anodes, and ester carbonate electrolytes, are taken as objects in these experiments, and the battery parameters shown in Table 1.

Table 1 Battery parameters

Parameter	Numerical value
The nominal voltage/V	3.6
Weight/g	47±0.5
The rated capacity/mah	3300
Charge cut-off voltage/V	4.2
Discharge cut-off voltage/V	2.5
Height/mm	65
Diameter/mm	18

The experiment subjects are prepared by the BattaTeq battery charging and discharging instrument. The preparation method and process for the lithium-ion batteries are as follows: firstly, the experiment subjects are discharged at a discharge rate of 0.5 C, until the voltage of the batteries are reduced to the discharge cut-off voltage of 2.5 V. Then, the batteries are left unaltered for 15 minutes. Next, charge the battery at a constant current rate of 0.5C until the voltage reaches 4.2V. Then conduct constant voltage charging at 4.2V until the current is below 0.05C. At this point, the state of charge of batteries reaches 100%. Finally, they are discharged to the expected SOC of 60%.Studies have shown that when the batteries reach 60%, they experience high levels of chemical energy and a high heat generation rate which increases the likelihood of battery thermal runaway occurring [25]. Charge and discharge are not carried out during the experiment.



a. Experimental device b. Single battery experimental

Fig. 1 Experimental system（1.water tank 2.filter 3.pressure relief valve 4.pump 5.damper 6.valve 7.flowmeter 8.pressure senor 9.explosion-proof box 10.nozzle 11.battery module 12.acquisition module 13.camera 14.computerdevice 15.test-bed 16.battery 17.heater 18.pressure gauge 19.strainer)

The experiment system is shown in Fig.1. The experiment system consists of a battery module, spray system module, and data acquisition module. In this study, using a heater triggers the thermal runaway of the battery. In the single cell experiment, products are collected by the device shown in Fig. 1(b). And in the battery module experiment, the heater and eight batteries are combined into a module and arranged into a 3 * 3 matrix. The battery and heating rod are tied together with high temperature-resistant tape to ensure heat transfer. The battery module is shown in Fig.1(a). The heating power is set to 160W. The spray system is located 10 cm above the battery module, and primarily includes a nozzle, water pump, thermostatic water tank, pressure gauges, and flow meters. The water in the tank is sucked in by the pump, then pushed through the pipe under pressure and discharged from the nozzle to cool the battery pack. The nozzle is a low-pressure atomizing nozzle and the diameter of the nozzle is 1.2mm. The results show that excessive spray flow has little effect on battery cooling [26]. Therefore, it is set to 0.2 MPa, 0.5 L min^{-1}. According to previous studies, this flow rate and pressure can effectively inhibit the thermal runaway of a single cell[25]. The spray angle of the nozzle is approximately 60°, and the size of the droplet is approximately

120μm. The whole experiment is carried out in an explosion-proof box. There is a camera set-up outside the explosion-proof box to record the process of thermal runaway. In addition to the camera, the data acquisition module also includes k-type thermocouples and NI equipment. The K-type thermocouples are set in the middle of the batteries' sidewall. The NI device is sampled at 100Hz.

2.2 Experimental procedure

This study primarily includes three experiments, namely, thermal runaway experiment of a single cell, thermal runaway propagation experiment of a battery module and thermal runaway propagation experiment of a battery module under a different spray time. When the battery temperature rise rate is higher than 1℃ per second, the heating rod is turned off.

In this experiment, the inhibition of thermal runaway propagation of batteries by spray is principally studied. Therefore, in the spray experiment, the spray is turned on when the battery temperature reaches its maximum and begins to drop. It is important to note is that the pump has been turned on prior to spraying and the pressure in the pipeline is stable to ensure smooth spraying after opening the valve.

2.4 Uncertainty analysis

The measurement accuracy of the measuring device is limited, so there are specific uncertainties in temperature, voltage and flow, and other data. The uncertainty of the instrument in measuring parameters is shown in Table 2.

Table 2 Component uncertainties of the experimental bench

Name	Uncertainty
The K type thermocouple	±0.75%
The rotameter	±4%
The stabilized voltage supply	±2%
The manometer	±0.4%
The current	±1%
The potential	±0.0015%

Additionally the uncertainty of the battery heating is ±49.9J and the uncertainty of the spray duration is ±0.1s.

3. Result and discussion

3.1 Single cell experiment

The thermal runaway process can be divided into three stages.

In the first stage, the battery temperature increases gradually. When the temperature reaches the decomposition temperature of the solid electrolyte interface, the anode is in contact with the electrolyte and reacts. Then the separator begins melting which causes an internal short circuit within the battery. The occurrence of an internal short circuit causes the voltage to drop sharply.

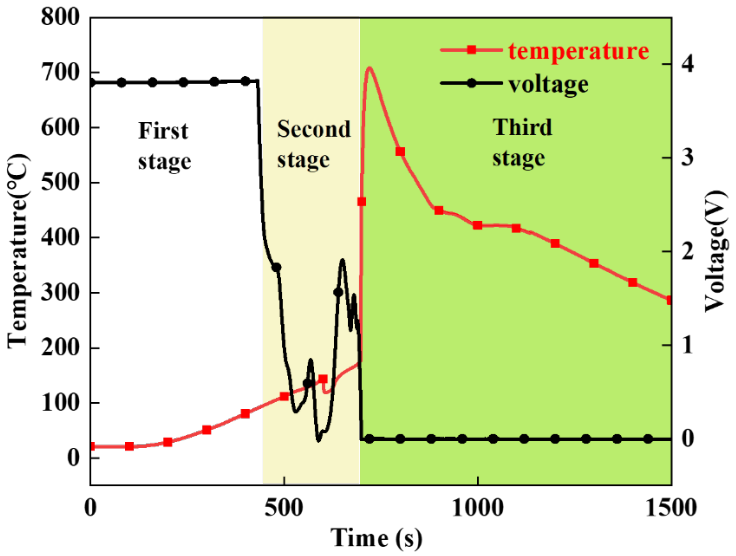


Fig. 2 Three stages of thermal runaway of an individual cell

The second stage is predominately the process of heat accumulation and gas release. An internal short circuit causes the temperature rise further. Simultaneously, the separator undergoes further decomposition. Finally, in the third stage, the positive and negative poles are directly connected, resulting in a large-scale short circuit. Simultaneously, the solvent and additives in the electrolyte would decompose on the negative electrode surface, and the metastable solid electrolyte interphase would regenerate. The decomposition and regeneration of the metastable solid electrolyte interphase causes voltage fluctuation.

Fig. 2 reveals that there has been a gradual increase in temperature and a steep drop in voltage between the first and second stages. As the temperature rises, the thermal runaway decomposition reaction rate in the battery is accelerated, and a large amount of gas is produced. and a large amount of gas would be produced. When the internal pressure of the battery reaches a certain value, the pressure valve will open, and the gas and electrolyte will be ejected. The gas contains flammable gas produced by the reaction, which will burn when it meets ignition conditions.

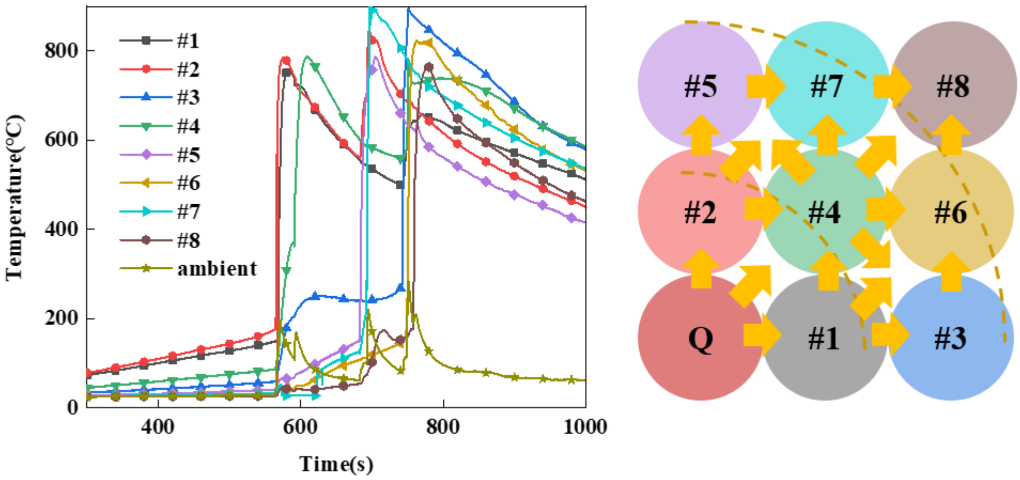
The significant phenomenon, occurring in the third stage, is the battery burning, and the temperature increases dramatically. During this process, the battery temperature rises rapidly and uncontrollably.

Table 3 Thermal runaway product mass

SOC of battery	Particle mass/g		Gas mass/g		Residual battery mass/g	
	Value	Average	Value	Average	Value	Average
60%	4.1	4.2	10.14	10.62	32.81	32.245
60%	4.3		11.1		31.68	
80%	7.93	7.745	12.15	11.995	26.95	27.275
80%	7.56		11.84		27.6	
100%	10.96	10.705	11.3	11.665	24.84	24.69
100%	10.45		12.03		24.54	

During the thermal runaway of the battery, large amounts of gas and solid particles are ejected from the safety valve. Table 3 shows that the particle mass loss increases approximately linearly with the increase of the stage of charge(SOC). This happens because the more energy a battery has, the more damage its thermal runaway will do to its internal structure. But the battery mass loss, caused by the gas, has a limited relationship with the battery SOC. This is primarily because at high temperature, the electrolyte vaporizes into a gas, and the mass of the electrolyte is determined. Both soot particles and gases would absorb and emit thermal radiation [27]. The maximum temperature can reach about 800°C when the battery jets flame. In this case, the effect of thermal radiation of particles and gas on thermal runaway propagation cannot be ignored, and further research is needed to verify its influence.

3.2 Thermal runaway propagation among batteries without spray cooling



a. Temperature in non-spray experiment b. Heat conduction and propagation paths

Fig. 3 Temperature in the non-spray experiment and thermal runaway propagation paths

Fig.3(a) presents the experimental data on the temperature of batteries in the non-spray experiment. Although #1 battery and #2 battery are both in direct contact with the heating rod, there is a time difference between them. A possible explanation for this might be that there are different contact thermal resistances in the assembly of battery modules, which makes a slight difference in the rate of heat conduction. The same phenomenon is also observed in the subsequent experiments.

Table 4 shows the time points and intervals of the thermal runaway of the battery. The propagation path of the battery can be inferred by the direction of heat transfer and the sequence of thermal runaway, as shown in Figure 3 (b).

The cells were divided into two groups by concentric circles centered on the heating rod. The first group consists of #1, #2, and #4 batteries. And the second group consists of #3, #5, #6, #7, and #8 battery. The interval between different groups is much longer than the interval within the group. The main reason for this is that the thermal resistance between different groups is greater. Interestingly, the time interval between the batteries in the second group is shorter. This is related to the flame produced by the thermal runaway of the first three batteries. The flame enhances thermal radiation and convection. This reduces the effect of heat conduction and reduces the time interval.

Table 4 Time point and interval of thermal runaway in non-spray experiment

Battery number	**#2**	**#1**	**#4**	**#5**	**#7**	**#3**	**#6**	**#8**
Time of occurrence/s	573	582	610	697	704	751	760	777
Interval time/s	0	9	28	87	7	47	8	17

3.3 Effects of emergency spray in propagation

The battery temperature for the five-second spray experiment and the continuous spray experiment are shown in Fig.5 and Fig.6. In the five-second spray experiment, #1 battery increases again after a brief drop in temperature under spray cooling, which is different from the temperature changes of #2 and #4 battery. The similar phenomenon is also found in the continuous spray. This may be because the internal structure of the battery is gradually damaged at high temperatures and an exothermic reaction occurs inside the battery. In the beginning, the damage inside the battery is minor, and the rate of the exothermic reaction is slow, and is less than the spray cooling rate. Under high temperatures, the degree of internal damage increases gradually, and the reaction rate also increases. Eventually, a violent internal reaction causes the battery to catch fire and burn, heating up dramatically. In the five-second spray experiment, the whole module finally has thermal runaway, and the five second spray could not completely inhibit the spread of thermal runaway. In the continuous spray experiment, except for the #1 battery and #2 battery, other batteries do not have thermal runaway. This proves that continuous spraying can effectively inhibit thermal runaway propagation.

It should be noted that when battery #1 caught fire in the continuous spray experiment, the spray was still coming out. When the fine water spray is sprayed onto the burning #1 battery, it vaporizes and fills the space. The water vapor absorbs the radiant heat emitted by the high-temperature cell and flame to the surrounding cells, which means that the water vapor weakens the radiant heat transfer process during thermal runaway propagation.

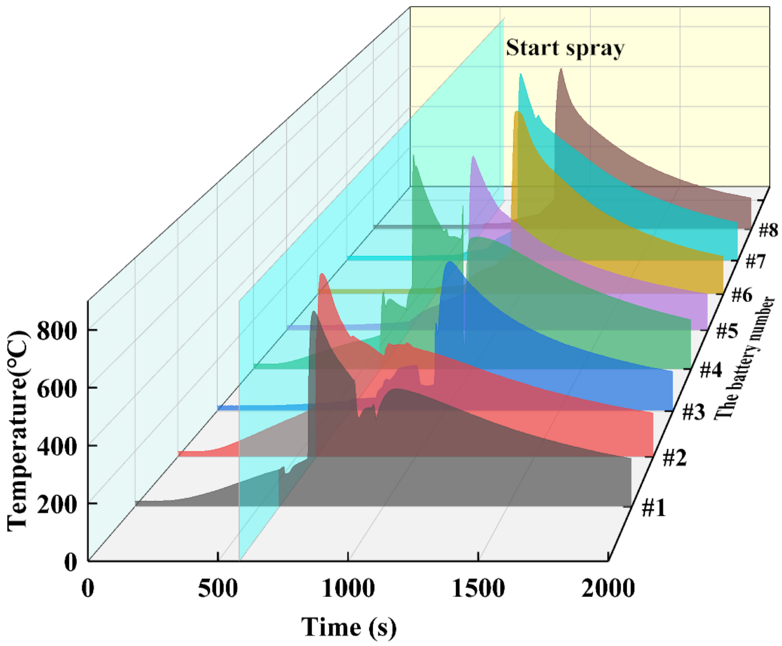


Fig. 4 Temperature in the five-second spray experiment

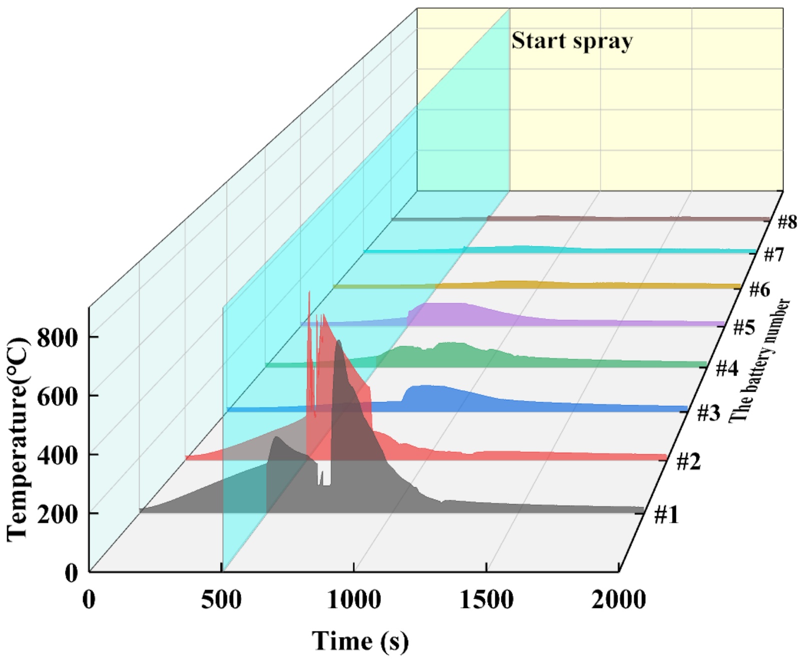


Fig. 5 Temperature in the continuous spray experiment

Table 5 Time interval between thermal runaway

The sequence of thermal runaway / Time interval(s)	1-2	2-3	3-4	4-5	5-6	6-7	7-8
Non-spray	9	28	87	7	47	8	17
5 second spray	115	11	156	3	95	10	26
Continuous spray	274						

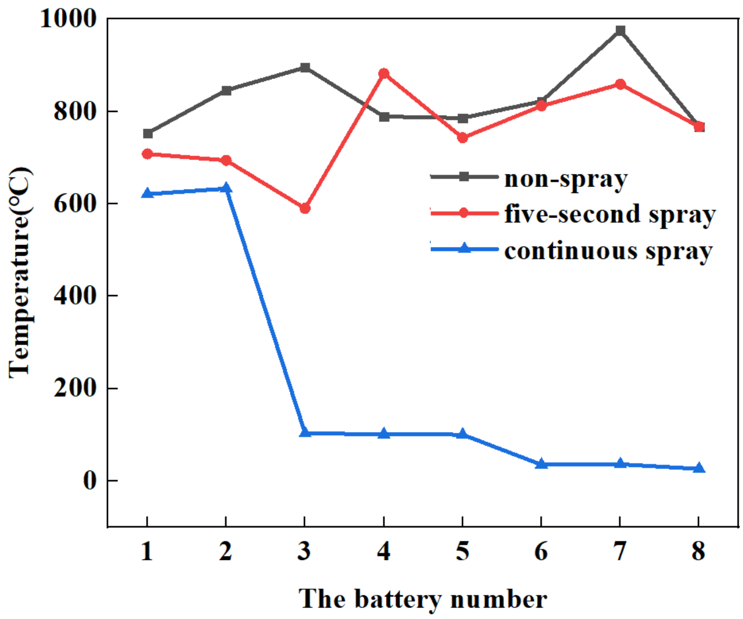


Fig. 6 battery maximum temperature

The maximum temperature of each battery in different experiments is shown in Figure 6. Due to the effect of spray, the maximum temperature of the first few batteries varies widely. Since the second experiment is only sprayed for five seconds, the gap between the maximum temperature of the battery and the non-spray experiment gradually narrows. It should be noted that the maximum temperature of the #4 cell in the five-second spray experiment is higher than the maximum temperature in the non-spray experiment. This may be because the #4 battery's side wall cracks in the non-spray experiment, taking some of the heat away. Additionally, the #6, #7, and #8 batteries were also observed to break the side of the battery during the five-second spray experiment. No similar phenomenon was observed in the continuous spraying experiment. The reason for this result is related to the manufacturing process of the battery. However, due to long-term preheating and spray, the metal surface becomes brittle, and consequently, are easy to break after fierce fire and burning. Therefore, the rupture primarily occurred in the last few batteries of the five-second spray experiment. The mass loss caused by battery rupture is even greater. The average maximum temperature of the batteries in the non-spray experiment is 828.75 °C.

The average maximum temperature of the batteries in the five-second experiment is

756.88 °C. In the continuous spray experiment, the average maximum temperature of the batteries is 206.8 °C. After five seconds of spray, the average maximum temperature of the battery dropped to only 71.87 ° C. It can be seen that the effect of short-term spray is not beneficial. It is obvious from Table 5 that the spray can delay the thermal runaway of battery ignition combustion in general. However, the time interval between the thermal runaway of two batteries in a row is shortened. The reason is related to the position of the battery and the preheating of the battery.

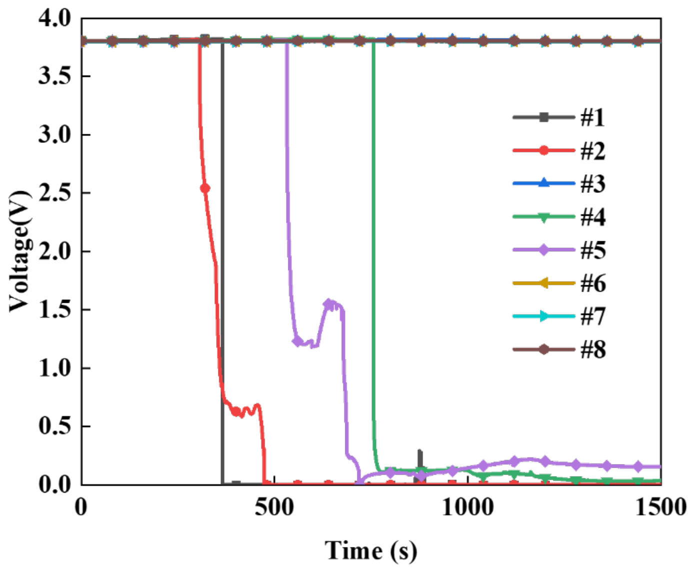


Fig. 7 Voltage in the continuous spray experiment

The voltage change of the battery in the continuous spray experiment is shown in Fig.7. It can be seen from Fig.5 and Fig.7 that the #1 and #2 batteries experience complete thermal runaway, and the batteries are completely damaged, which are called damaged batteries (DBs). It is worth noting that although only #1 and #2 batteries are on fire, the voltage of batteries #4 and #5 actually also drops to 0V. These batteries, which have failed and are no longer usable, are called failed batteries (FBs). The remaining batteries are called survived batteries(SBs) and these are batteries which have not changed significantly but have been sprayed with water. As can be seen from Table 6, the quality of the FBs also changes significantly in the continuous spray experiment. According to this data, we can infer that the loss of FBs' mass is related to voltage changes. This phenomenon may be caused by the internal short circuit of the battery, which generates some heat and causes the separator to melt. The gas produced by the internal reaction is released from the battery, resulting in a loss of battery mass. But these two batteries do not catch fire. It can be observed that, although the internal short circuit is the same feature of battery thermal runaway under different trigger modes [28], it is not the immediate cause of thermal runaway. Comparing the temperature of the batteries in Figure 7 finds that the temperature does not change significantly even if a short circuit occurs inside the battery. It follows that a short circuit inside the battery does not generate much heat and is not the key factor that causes the battery to catch fire. This finding is consistent with that of Ren, who showed that heat generated by an internal short circuit does not trigger a thermal runaway [29].

Table 6 Battery experiment results under continuous spray

Number	Significant drop in voltage	Burning	Quality loss
#1 (⊡)	✓	✓	13.69g
#2 (⊡)	✓	✓	15.59g
#3 (⊗)	×	×	0.27 g
#4 (Δ)	✓	×	19.61g
#5 (Δ)	✓	×	16.02g
#6 (⊗)	×	×	0.19g
#7 (⊗)	×	×	0.13g
#8 (⊗)	×	×	0.15g

Note: **(⊡)**: The damaged battery experienced a complete thermal runaway. **(Δ)**: The failed battery voltage is abnormal and cannot be used. **(⊗)**: The battery is sprayed, but there is no significant change.

The SBs can still be charged and discharged. But after standing for several days, it is found that the voltage of SBs drops to zero and they become FBs. This may be caused by water mist and water vapor that liquefies on the surface of the cell during the continuous spray condition, which penetrates the cells and causes reactions that diminish the performance of the cells. Even if these batteries can no longer be used, they can still be triggered into thermal runaway.

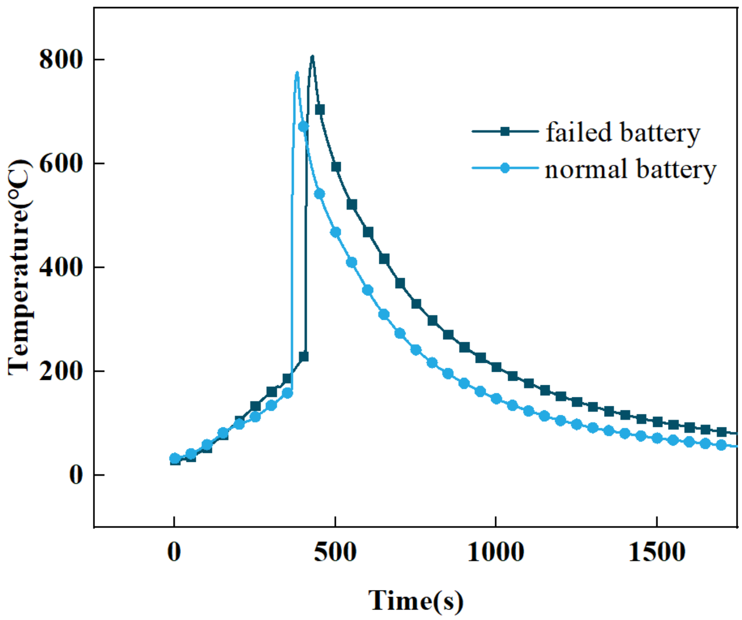
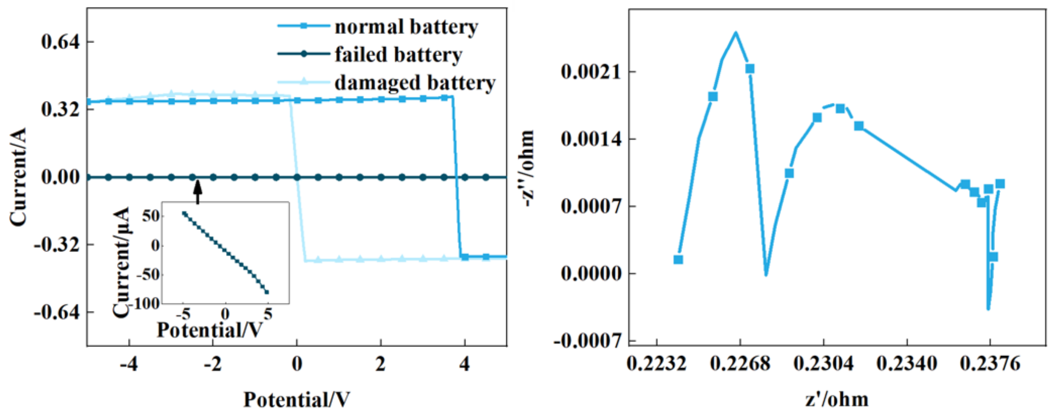


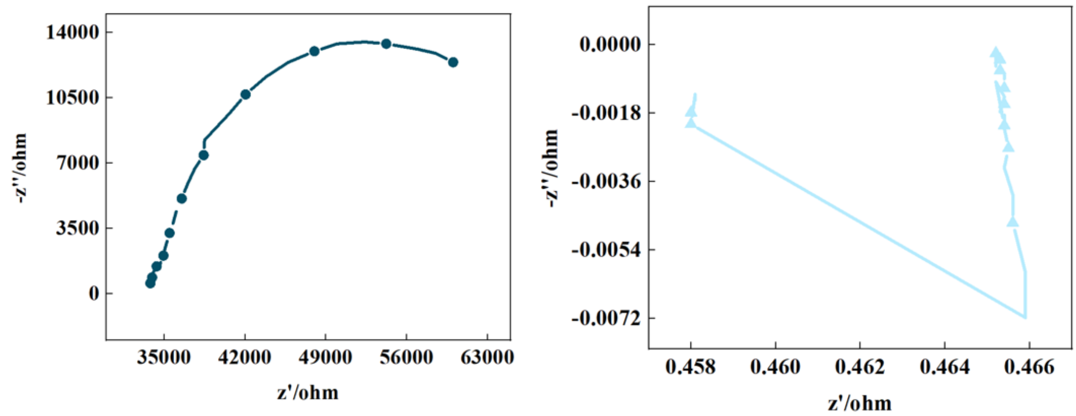
Fig. 8 Temperature of normal and failed batteries

The trigger temperature of the thermal runaway of the normal battery is about 160°C. However, when the battery voltage drops to 0V, its trigger temperature can reach 230°C at most, as shown in Figure 8. Also, the safety valve of a failed battery opens at a higher temperature than a normal battery. The reason for this may be that the battery has lost vast amounts of energy and needs more external energy input to trigger thermal runaway.

The internal resistance of the battery is measured by the voltammetry of CHI660E. From the voltage and current in Figure 9(a), it can be identified that the internal resistance of these batteries is thousands of ohms or even tens of thousands of ohms. In fact, the internal resistance of a normal cell is about a few micro Ohms. This may be because the internal structure is destroyed, which leads to a sharp increase in internal resistance and leads to an internal short circuit.



a. Battery internal resistance by voltammetry b. EIS of normal battery



c. EIS of failed battery d. EIS of damaged battery

Fig. 9 Impedance of Battery

The electrochemical impedance spectroscopy of these batteries is shown in Figure 9(b)(c)(d). The real and imaginary parts of the resistance of the failed battery are much greater than those of the normal battery. There are two arcs in the EIS of the normal battery, but only one in EIS of the failed battery and the damaged battery. In addition, the real part of the resistance of the damaged battery is slightly higher than that of the normal battery. The sign of the imaginary part of the impedance of the damaged battery is the opposite of the normal sign.

3.4 The cooling capacity required to suppress the thermal runaway

A large amount of heat is generated during thermal runaway of the battery. The heat predominately comes from the internal short circuit and chemical reaction of internal material. We assume all of the things that are involved in the reaction are completely reactive. Internal material parameters are shown in the following table.

Table 6 Mass and enthalpy of battery reactants

Parameter	Anode	Cathode	Separator	Electrolyte	SEI
mass(g)	4.7	17.04	0.7	4.1	0.75
enthalpy(J/g)	1714[30]	218[31]	-190[30]	800[30]	257[32]

The enthalpy of a substance is related to the energy it contains. In the absence of flow work, the change in enthalpy is the change in internal energy or heat. Assuming that all the energy of a substance is converted into heat, the heat production reaches its maximum when the reaction of the substance is complete. The maximum heat produced by each part can be obtained by the following formula:

$$Q_{max,c} = m_c * H_c \tag{1}$$

where m_c is the mass of a part of battery, and H_c is the enthalpy of reaction for this part, and $Q_{max,c}$ is the maximum heat production of this part.

And the total electrical energy of the battery can be calculated by the following formula:

$$Q_i = C * V \tag{2}$$

where C is the capacity of the battery, and V is the voltage corresponding to this capacity, and Q_i is the total electrical energy.

In the event of a short circuit, 70% energy of the battery mentioned earlier will be released within 60 seconds, causing the battery temperature to rise [33]. The heat generated by the short circuit of the battery can be calculated using the following equation:

$$Q = \eta * Q_i \tag{3}$$

where Q represents the short-circuit heat release of the battery, and η is the heat conversion coefficient [34], which is the ratio of the internal energy of the battery to heat energy.

Thus, the greater the voltage, the greater the short-circuit energy. When the SOC of the battery is 60%, the voltage is 3.8V, and the short circuit energy of the battery is 18960.48 J. Finally, the heat yield of 60% SOC battery is 34070.75J.

When the battery has thermal runaway, the spray is turned on to produce water mist, which absorbs heat and vaporizes. The heat absorption of the whole process can be calculated from the following equation.

$$Q_w(T) = \begin{cases} C_p \cdot q_m \cdot (T_b - T_0), (T \leq T_b) \\ C_p \cdot q_m \cdot (T_b - T_0) + h_f \cdot m, (T \geq T_b) \end{cases} \tag{4}$$

where C_p is the specific heat capacity of water, T_b is the evaporation temperature of water, T_0 is the environment temperature, and q_m is the mass flow rate of the water. The maximum cooling capacity provided by the spray is 21.42kW. Assume that the spray is sprayed equally in all directions. The proportional coefficient of the cooling capacity of a battery can be obtained from the formula below:

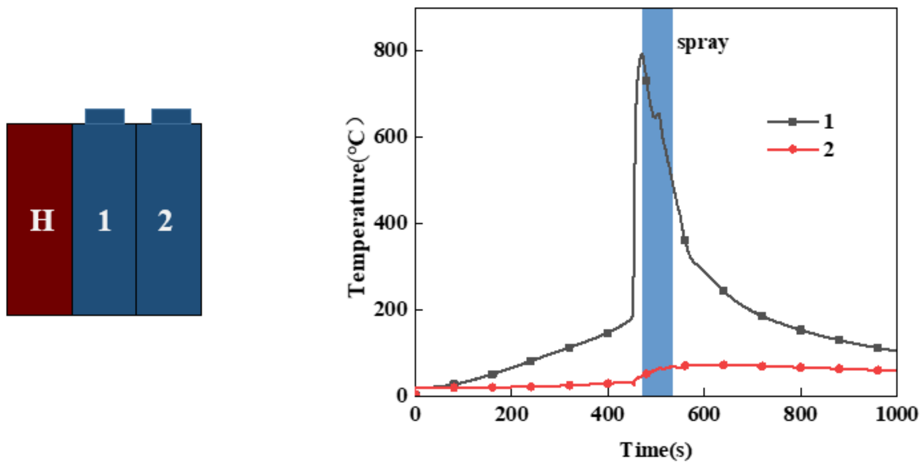
$$\alpha = \frac{r^2}{\left(h \cdot \tan\frac{\theta}{2}\right)^2} \tag{5}$$

where r is the radius of the battery, h is the distance between battery module and nozzle, and θ is the jet angle of nozzle.

The maximum cooling capacity of a single battery can be obtained from the formula below:

$$Q_c = \alpha \cdot Q_w(T) \tag{6}$$

The maximum value of Q_c is 520.05W.



a. The module of heating rod and battery b. 65s spray experiment

Fig. 10 Verification experiment

When thermal runaway occurs in a single cell within a battery pack, the energy generated by that cell is sufficient to induce thermal runaway in surrounding cells. To inhibit the propagation of thermal runaway, spray is employed to eliminate the heat generated by the thermal runaway of the cell. As the calculations show, under ideal conditions, 65 s of spray cooling can eliminate the heat generated by the thermal runaway process of the cell. The experimental setup is simplified as shown in Figure 10(a). The spray was activated and continued for 65s when the first cell underwent thermal runaway. The experimental results show that the 65 s spray does not eliminate all the heat generated by the battery. This may be due to the fact that the fine water spray evaporates into steam after contact with the hot battery, which affects the contact of subsequent fine water sprays with the battery, thus leading to a decrease in the cooling effect. The spray significantly reduced the heat transfer between the cells, resulting in a slow increase in the temperature of the second cell which eventually remained stable. Thermal runaway propagation between cells is suppressed with 65 seconds of continuous spraying. In practice, it is not necessary to completely eliminate the heat, just enough to keep the heat from triggering other batteries to initiate thermal runaway. In the battery pack, as the battery generates heat, it is also transferring heat to the surrounding battery and the spray also has a cooling effect on other batteries that are not out of control, which makes the situation more complicated. This requires further investigation and calculations.

4 Conclusion

Lithium-ion batteries are widely used in electric vehicles. In a number of vehicles, including those produced by Tesla, the power is provided by thousands of 18650 cylindrical batteries arranged tightly in battery packs. When a single battery experience thermal runaway, it often triggers thermal runaway of all the batteries in the battery pack, causing the electric vehicle to spontaneously ignite. Therefore, when a battery experiences thermal runaway, a solution to inhibit its propagation has attracted increasing attention. In this study, the thermal runaway of a single battery and a battery module has been investigated and the influence of spray time on thermal runaway propagation explored. The study identifies the following conclusions :

1. A large amount of gas and solid particles will be produced in the process of battery thermal runaway. With the increase of battery SOC, the mass of particles collected after thermal runaway will increase, and the mass loss before and after battery thermal runaway will also gradually increase. The increase of solid

particles leads to the increase of radiant heat, which is likely to promote thermal runaway propagation.

2. According to the time interval of thermal runaway and the direction of heat transfer, it can be inferred that the thermal runaway of the battery is transmitted outward in concentric circles. Batteries on the same concentric circle are a group. The interval time of thermal runaway within the group is far less than that between groups.
3. If the spray lasts for a short time, some batteries will reignite, so the thermal runaway cannot be effectively suppressed. However, the five-second spray duration increases the time between thermal runaway transmission by approximately 4 minutes, which can help increase the chances of escape. With continuous spraying, the thermal runaway propagation of the battery can be inhibited. Only the two batteries closest to the heating rod will burn. Some batteries do not burn, but they do fail. The other spray-on batteries have no obvious change, but after standing for a long time, the voltage of other batteries also drops to 0V, which may be related to the corrosion of the battery structure by water. These batteries have extremely high internal resistance. Moreover, the opening temperature of the safety valve and the triggering temperature of thermal runaway of these batteries have increased, and more external energy input is required to trigger the thermal runaway.
4. According to the mass and enthalpy of reactants and the short circuit energy, the maximum heat generated by a 60% SOC lithium-ion battery during thermal runaway is estimated to be 34070.75J. The energy generated by the battery is enough to trigger the thermal runaway of the surrounding battery. It takes 65s to eliminate the maximum heat production of the battery when only the ideal spray cooling is considered. It is found that the thermal runaway propagation is effectively inhibited by the simplified experiment

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